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NILKAMAL SCHOOL OF MATHEMATICS, APPLIED STATISTICS & ANALYTICS

Deep Learning for Portfolio Optimization

GROUP 8: SEMESTER III

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Abstract and Problem Statement

Abstract

- Start from the reference LSTM portfolio model (Zhang, Zohren & Roberts, 2020) and test it against 7 benchmarks.
- Compare LSTM, Attention, LSTM+EIIE and Attention+EIIE under one identical protocol; winner chosen on validation Sharpe only.
- Plain LSTM wins: validation Sharpe 2.56 and highest test Sharpe in all 3 experiments.
- Same 4 models retrained on Indian ETFs: Attention wins Exp 1–2, LSTM wins Exp 3.

Problem statement

Can a network read raw market history and output portfolio weights directly, trained to maximise the Sharpe ratio?

- Classical (Markowitz) methods rely on unstable return and correlation estimates.
- Diversification breaks down in crises; normal-return assumptions miss fat tails.
- Predicting returns well ≠ building the best portfolio.

Old: predict prices → optimise

New: market data → weights

We test if attention and EIIE improve a Sharpe-trained LSTM, and if it works in India.

Aim and Objectives

Aim

To evaluate, under a single controlled protocol, whether attention and EIIE improve a Sharpe-trained LSTM portfolio model, and which of the four models works best on Indian ETFs.

1

Baseline LSTM

Re-implement the reference LSTM and test it against 7 independently computed benchmarks.

2

Comparison

Compare LSTM, Attention, LSTM+EIIE and Attention+EIIE under identical conditions.

3

India

Retrain all four models on Indian ETFs; name the winner of each experiment.

4

Transparency

Record every deviation and caveat; no number quoted without its limits.

Dataset

Asset class	US ETFs (Part 1)	Indian ETFs (Part 2)
Stock	VTI	NIFTYBEES.NS
Bond	AGG	LTGILTBEES.NS
Commodity	DBC	GOLDBEES.NS
Volatility	^VIX	^INDIAVIX

Annualised	VTI	AGG	DBC	^VIX
Mean return	0.101	0.045	-0.030	0.825
Std deviation	0.203	0.051	0.194	1.287

US daily returns: very different risk across the 4 asset classes

US (Part 1)

- 7 Feb 2006 – 29 Apr 2020: 3,581 trading days.
- Retrain 2011, 2013, 2015, 2017, 2019; test to Apr 2020.

India (Part 2)

- 21 Jun 2019 – 30 Oct 2026: 1,776 days.
- Test: 3 Jan 2022 – 30 Oct 2026.

Source: Yahoo Finance (yfinance). Input = 50 days × 8 = 400 features; z-score fitted only on past data.

Dataset

Date	AGG_Adj Close	DBC_Adj Close	VTI_Adj Close	VIX_Adj Close	AGG_Return	DBC_Return	VTI_Return	VIX_Return
07-02-2006	52.51906586	18.63015747	43.28556442	13.59000015				
08-02-2006	52.49286652	18.55088234	43.59696198	12.82999992	-0.000498854	-0.004255205	0.007194028	-0.055923489
09-02-2006	52.52432251	18.7252903	43.51392746	13.11999989	0.000599243	0.009401599	-0.001904594	0.022603271
10-02-2006	52.41420746	18.3843956	43.60388947	12.86999989	-0.002096458	-0.018205042	0.00206743	-0.019054878
13-02-2006	52.45088959	18.09900284	43.40665054	13.35000038	0.000699851	-0.015523641	-0.004523425	0.037296076
14-02-2006	52.36699677	17.94837379	43.81840897	12.25	-0.001599455	-0.008322505	0.009486068	-0.08239703
15-02-2006	52.42469025	17.68675804	43.98448944	12.31000042	0.001101715	-0.014576014	0.003790199	0.004897993
16-02-2006	52.37224197	17.90080643	44.32704544	11.47999954	-0.00100045	0.012102183	0.007788109	-0.067424927

Methodology and Pipeline

1

Feature stacking

Stack 50 days × 4 assets (price + return) into one 50×8 input block.

→

2

Neural encoding

Model (LSTM or Attention, ± EIIE) reads the window and outputs one raw score per asset.

→

3

Softmax

Scores become long-only weights in [0,1] that sum to 1.

→

4

Return realisation

Portfolio return = weights × actual returns − transaction costs (optional volatility scaling).

→

5

Sharpe optimisation

Loss = −Sharpe; backpropagation updates the model to raise it.

Same data, loss and optimiser for all 4 models

Adam · batch 64 · 100 epochs

Validation: last 10% (chronological)

3 seeds (0, 1, 2); results = mean

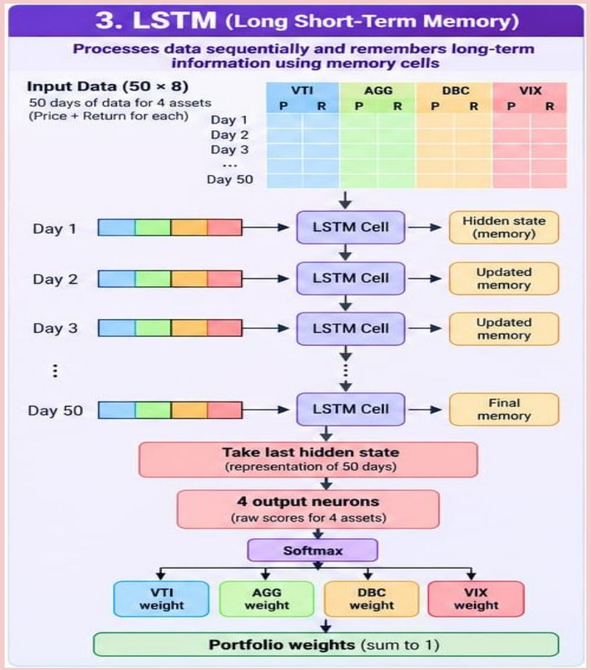
Retrain every 2 years, expanding window

Winner picked on validation; test used once

Fair comparison: only the model architecture changes; everything else is held fixed.

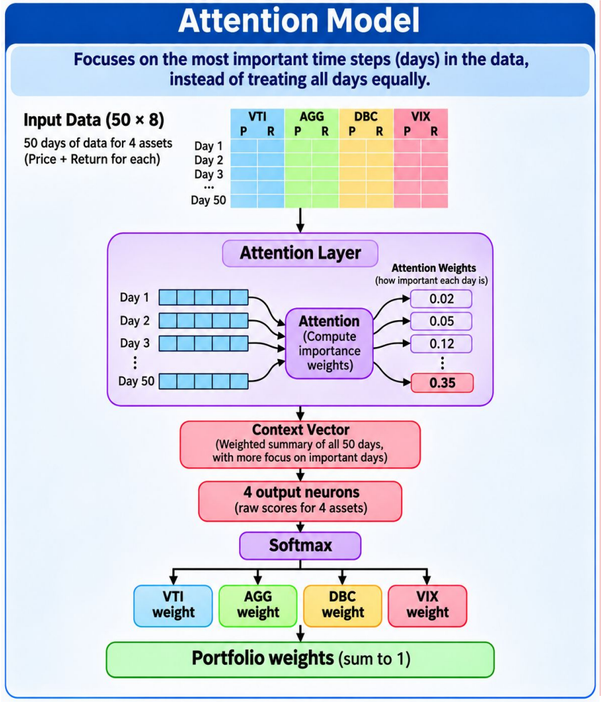
Models Overview: LSTM and Attention

A · LSTM	B · Attention	C · LSTM + EIIE	D · Attention + EIIE



A · LSTM: reads day by day

- Processes the 50 days one at a time, updating a memory (hidden state).
- Gates decide what to keep or forget, so long-term patterns survive.
- Last hidden state summarises 50 days → 4 scores → softmax weights.



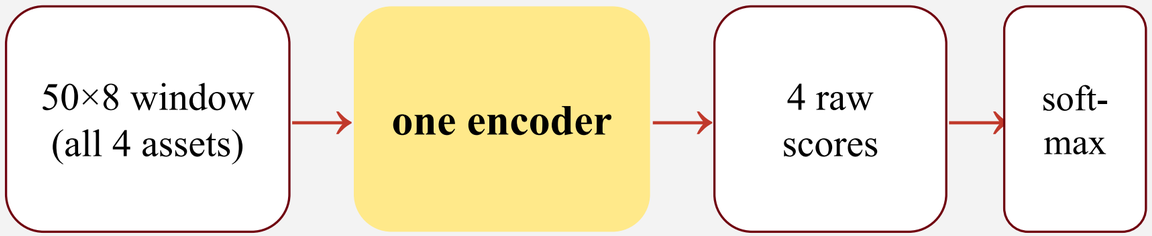
B · Attention: looks at all days at once

- Every day attends to every other day and learns which past days matter most.
- Query, Key, Value
- Pool compresses everything into one summary, then softmax turns it into portfolio weights .
- Positional Encoding

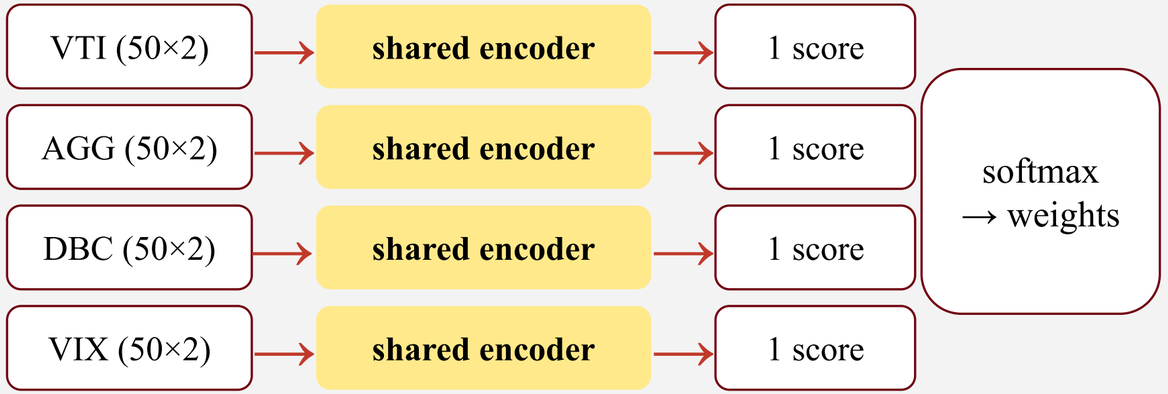
LSTM = sequential memory (compact). Attention = global comparison (heavy).

Why EIIE helps :

Joint input (A, B)



EIIE (C, D): one shared evaluator per asset



Shared weights

Same mini-network for every asset

Consistent scoring

Every asset is evaluated by the same rule, so scores are directly comparable.

Scalable

Assets can be added or removed without changing the architecture.

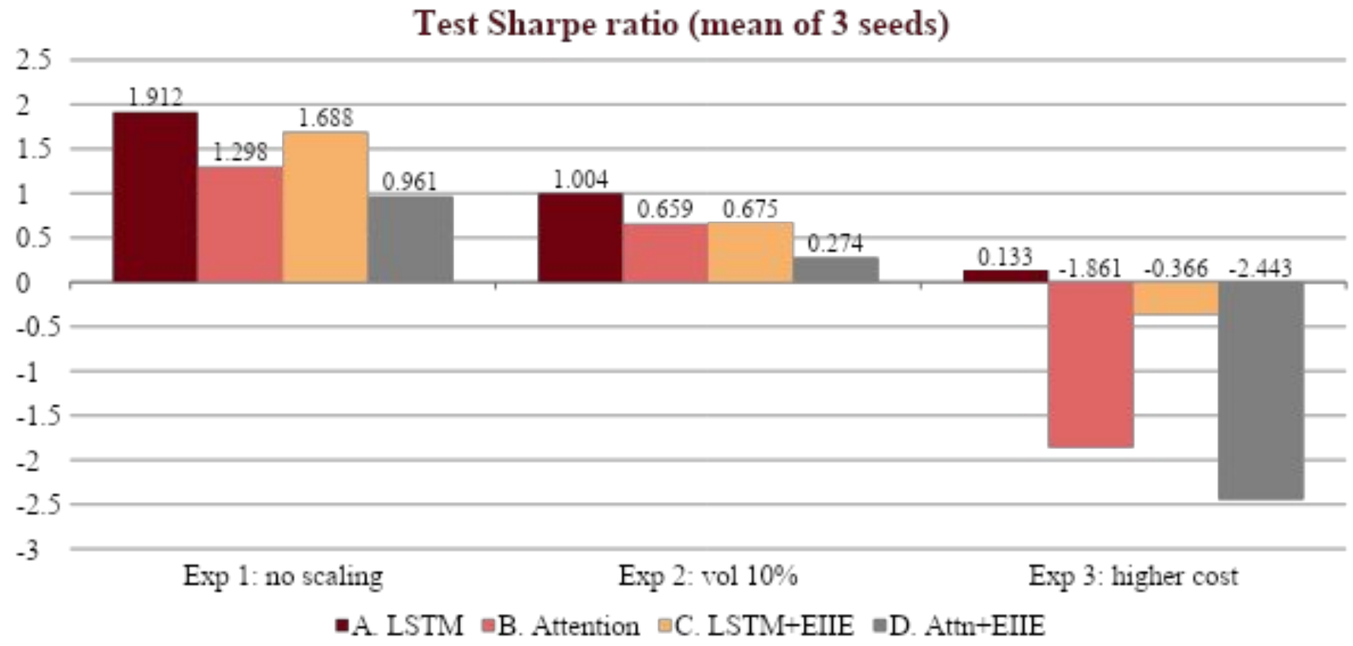
Independent streams

Each stream sees only its own asset; streams interact only at the softmax.

EIIE: Ensemble of Identical Independent Evaluators

Results: Four Models (US Data)

Validation	A. LSTM	B. Attn	C. LSTM+E	D. Attn+E
Val. Sharpe	**2.562**	1.775	2.051	0.913



Winner: A · LSTM

- Highest validation Sharpe (2.562): chosen before any test number.
- Highest test Sharpe in all 3 experiments.
- Smallest model (**19,204 parameters**) yet best.

Experiments

Exp 1: no vol scaling, C = 0.01%

Exp 2: vol target 10%, C = 0.01%

Exp 3: vol target 10%, C = 0.1%

Takeaway: Plain LSTM wins on validation (2.562) and on test Sharpe in 3 of 3 experiments.

Why the LSTM Won

1 · Attention overfits

- ≈ **21× the parameters**: it memorises the training period.
- Train Sharpe 19.7 vs validation 1.8 (B).

2 · Attention over-trades

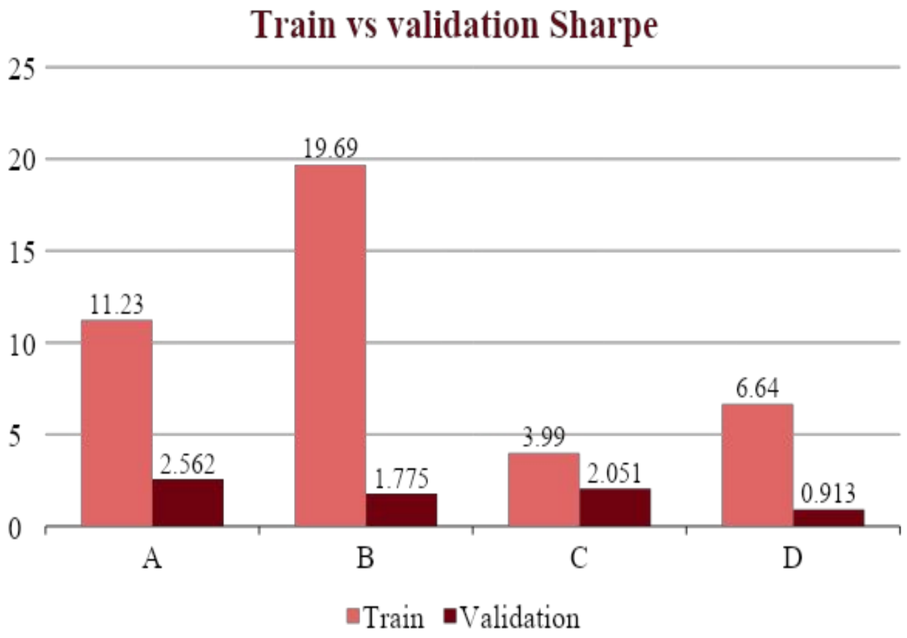
- Increased Transactions Cost
- **Costs *10 -** (Exp 3): B −1.86, D −2.44, A +0.13.
- Daily turnover: A 0.19 vs B 0.58, D 0.97.

3 · EIIE takes too much risk

- Greater Risk without generating proportionately higher returns
- Std 0.55 (C) vs 0.125 (A); MDD 0.35 vs 0.17.
- C holds only 4% stock and 4% bonds.

4 · LSTM is balanced

- Holds 51% stock, 22% bonds, 18% commodity, 9% volatility.
- Matches the paper: Sharpe 1.912 vs 1.858.
- Helps maintain balance between Returns, Risk and trading cost.



Takeaway: The LSTM generalises better, trades less and maintained better risk control than the alternative model we tested.

Indian Dataset: Training the 4 Models on Indian ETFs

Model	Params	Val. Sharpe	Seed sd
A. LSTM	19,204	1.564	1.854
B. Attention	14,788	-0.049	0.141
C. LSTM + EIIE	17,473	0.307	1.273
D. Attn + EIIE	14,497	0.498	1.409

Setup (same protocol as the US)

- 4 models × 3 seeds × 3 windows (2022, 2024, 2026) = 36 fits.
- Test: 3 Jan 2022 – 30 Oct 2026 (1,164 days).

Validation-selected winner: A · LSTM

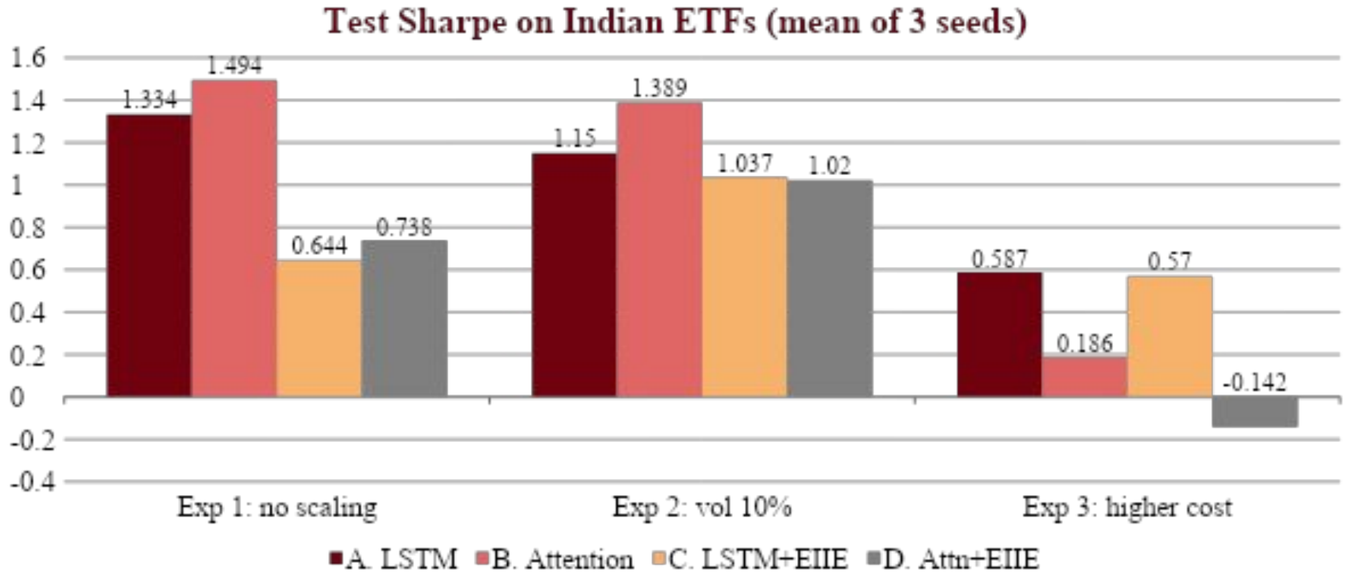
- Highest validation Sharpe (1.56) on the first window.
- Core mix: 71% Nifty ETF + 22% gilts(government securities); lowest turnover (0.13).

Avg weights	Nifty	Gilt	Gold	VIX	Turn-over
A. LSTM	0.71	0.22	0.03	0.04	0.13
B. Attn	0.68	0.29	0.02	0.02	0.30
C. LSTM+E	0.23	0.09	0.09	0.58	0.24
D. Attn+E	0.13	0.21	0.08	0.58	0.53

Takeaway: EIIE models put ~58% in India VIX

Sharpe = return per unit of risk; seed = a repeat of training with a different random start; turnover = how much the weights change daily.

Results: Four Models on Indian ETFs



Model A Sharpe	Exp 1	Exp 2	Exp 3
US (2011–Apr 2020)	1.912	1.004	0.133
India (2022–Oct 2026)	1.334	1.150	0.587
India experiment winner	**B 1.494**	**B 1.389**	**A 0.587**

Reading the results

- Low cost (Exp 1–2): Attention does best (1.49, 1.39).
- High cost (Exp 3): LSTM does best (0.59).
- Why: Attention trades a lot, so higher costs hurt it more (1.39 → 0.19)
- EIIE: It wins nothing. It puts most of its money in India VIX, which you can't actually trade.

Takeaway: India: Attention leads at low cost, LSTM at high cost; validation-selected LSTM.

Conclusion

1 Baseline LSTM

LSTM without volatility scaling has highest (Sharpe 1.912); but does not give highest Sharpe when volatility scaling is done.

2 Model comparison (US)

Plain LSTM is best in all 3 experiments and on validation (2.56). Attention and EIIE both did perform well. Attention was 21× bigger and memorised the training data.

3 India

LSTM is validation-selected. Attention wins Exp 1–2 (Sharpe: 1.49, 1.39), LSTM wins Exp 3 (Sharpe: 0.59).

4 Limits and next steps

The Indian dataset was much smaller than the US one: 1,777 days versus 3,581. This is because Indian ETFs are newer and the early data for gilt ETFs is sparse. As a result, we had only 56 validation samples and 3 seeds.

LSTM best in 3 of 3 US experiments | India: Attention 2, LSTM 1

THANK YOU!

Open for Questions..